

# Suhan Roy

Portfolio: [wsl.iitb.ac.in/suhan-roy](https://wsl.iitb.ac.in/suhan-roy)

Github: [github.com/suha-ain](https://github.com/suha-ain)

Email: [suhanski11@gmail.com](mailto:suhanski11@gmail.com)

Mobile: +91-7044766369

## EDUCATION

- **International Institute of Information Technology Bangalore** Bangalore, India  
*Master of Science by Research – Data Science and Artificial Intelligence; GPA: 3.35/4.0 July 2024 – June 2026*
  - **Relevant Coursework:** Machine Learning, Optimization, Multi-Agent Systems, Reinforcement Learning, NLP
- **Ramkrishna Mahato Government Engineering College** Purulia, India  
*Bachelor of Technology – Electronics and Communication Engineering; GPA: 7.47/10 July 2016 – June 2020*
  - **Relevant Coursework:** Data Structures, Algorithms, AI, ML, OOP, DSP

## SKILLS SUMMARY

- **Languages:** Python, C++, SQL, Java, Julia
- **Frameworks:** Scikit-learn, PyTorch, TensorFlow, Keras, SpaCy
- **Tools:** Docker, Kubernetes, Git, Ansible, Selenium, MySQL
- **Platforms:** Linux, Windows, AWS, GCP, Huggingface
- **Soft Skills:** Leadership, Event Management, Writing, Public Speaking, Time Management

## WORK EXPERIENCE

- **Datalyca Technologies** Remote  
*Research Intern June 2025 – Present*
  - **Decision Transformer RL:** I have built a badminton playing model with offline Reinforcement Learning from tournament matches data using Decision Transformers that can predict shots to play that maximises player's chances to win rallies and get maximum points.
  - **3D Visualization:** I'm now using 3D human poses rendering models to accurately build 3D videos for professional badminton coaching
- **Web Sciences Lab, IIIT Bangalore** Bangalore, India  
*Graduate Research Assistant, Teaching Assistant July 2024 – Present*  
Supervisor: Dr Sushree Behera, Dr Srinath Srinivasa
  - **Navigated Learning Project:** I am working in the Navigated Learning Project funded by Gooru Labs on Deep Reinforcement Learning for Active Learning Maps. We are building a learning Navigator for Personalised courses on E learning platforms that generates unique learning pathways based on competencies and assimilations
- **Indraprastha Institute of Information Technology Delhi (IIIT-D)** New Delhi, India  
*Research Assistant July 2023 – April 2024*  
Supervisor: Dr A V Subramanyam
  - **Vision-Language Models:** Worked on text-to-person Re-ID using CLIP and BLIP-based models.
  - **Masked Image Modeling:** Implemented Masked Image Modelling Inspired by SimMIM paper from Microsoft on CLIP based Person Reidentification architectures like IRRa and have achieved a 0.15 percent increase in accuracy over the latest works in this area
  - **Segmentation Pretraining:** Explored DDPM models for segmentation with ViT backbones for Image pre training.
- **Capgemini Engineering** Gurugram, India  
*Analyst Jan 2021 – June 2023*
  - **Protocol Testing:** Developed C++ and TTCN-3 code for VoLTE, SIP, ARP, BGP, and OSPF testing.
  - **Automation Tools:** Built test automation systems using Selenium, Robot, and Ansible.
  - **Traffic Simulation:** Wrote Python/TCL test scripts for simulating traffic with L2 protocols.

## PROJECTS

- **AdEMAMix Optimizer in Julia:** I implemented the AdEMAMix optimization algorithm from scratch in Julia and used it to train Vision Transformer and ResNet18. Did a comparison study with AdEMAMix, Adam and RMSProp variants on various datasets including CIFAR10 revealing key insights into the interplay between neural architectures and optimization algorithms.
- **Multi-Agent Emotional Modeling:** Using an Agentic AI framework and architecture to study interactions of agents with personalities and emotions in a complex Multi Agent System and developing various situations and games with Psychological Game Theory to study outcome and emergent properties of the system.

## HONORS AND AWARDS

- **Competitive Programming:** Achieved Global Rank 1955 in CodeChef August Lunchtime 2020.
- **Campus Leadership:** Executive Team Member, CodeChef Campus Chapter.

## VOLUNTEER EXPERIENCE

- **ICVGIP 2024, IIIT Bangalore** Bangalore, India  
*Student Volunteer Jan 2024*
  - **Conference Logistics:** Supported organization and hosting of ICVGIP 2024.
- **Miles4Meals, IIIT Bangalore** Bangalore, India  
*Event Organizer Feb 2024*
  - **Community Engagement:** Organized the flagship 10K Run with the Sports Committee.